

Learning objectives

In this lesson you will learn to:

- define the terms 'nibble' and 'byte'
- convert between denary and 8-bit binary numbers.

For more detail on this topic, and additional student activities, refer to Topic 2 of the student book.

Numbers

Previously, you learned that all data is represented using binary in a digital computer.

One type of data that needs to be represented is numerical data.

You will be looking at a particular set of numbers: **unsigned integers**.

This means whole, positive numbers (more than or equal to 0).

Numbers

Humans count using **denary** numbers (base 10).

We use 10 symbols: 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

This is because we have ten fingers.

You already know how denary numbers work.

10^3	10^2	10^1	10^0	
1000	100	10	1	
5	0	4	9	
$(5 \times 1000 =$ 5000) 5000	$(0 \times 100 =$ 0) + 0	$(4 \times 10 =$ 40) + 40	$(9 \times 1 = 9)$ + 9	= 5049

Resources

Computers count using binary numbers (base 2).

They use just 2 symbols: 0 and 1

2^3	2^2	2^1	2^0	
8	4	2	1	
1	1	0	1	
(1 x 8 = 8) 8	(1 x 4 = 4) + 4	(0 x 2 = 0) + 0	(1 x 1 = 1) + 1	= 13

Counting

Let's count in binary together:

1111

Remember 2^n where $n = 4$, gives us 16 unique values, 0 through 15

Nibbles

A set of 4 **bits** (binary digits) is called a **nibble**.

A single nibble can store 16 unique values (2^4) – the denary numbers, 0 to 15.

We can use a table to help us remember the **place value** of each column.

8	4	2	1
0	1	1	0

To convert this binary number back to denary, we can add the column values which contain the digit 1.

So the bit pattern 0110 is the same as $4 + 2$, which equals 6.

Nibbles

You can also convert from a denary number into its binary equivalent.

You can use the same table to help you. However, instead of adding, you'll subtract.

For example, to convert 13 in denary into binary:

- Largest column value that fits: 8
- Put a 1 in the 8 column
- $13 - 8 = 5$; move along a column
- $5 - 4 = 1$
- Put a 1 in the 4 column; nothing left.
- 13 in denary = 1101 in binary.

8	4	2	1
1	1	0	1

Bytes

Two nibbles make a **byte**. A byte is 8 binary digits.

A byte is the standard unit of measurement in digital computing.

A byte can represent 256 values (0 to 255 in denary).

Converting uses the same process as with a nibble, but with more columns.

128	64	32	16	8	4	2	1
1	0	0	0	0	1	1	0

So this binary number converted to denary would be: $128 + 4 + 2 = 134$.

Most significant bit

It is important to remember that all binary patterns are written from the largest value to the smallest value, as shown in the table.

Exactly the same approach is used for the normal denary numbers that we use every day.

128	64	32	16	8	4	2	1
1	0	0	0	1	1	0	0

In binary numbers we use the term **most significant bit (MSB)** to refer to the leftmost bit in any binary number.

This means that the leftmost digit has the largest place value.

Wrap up: you have learned how to...

- ✓ Define the terms 'nibble' and 'byte'.
 - A byte is 8 bits. A nibble is half a byte or 4 bits.
- ✓ Convert between denary and 8-bit binary numbers.
 - Adding place values for binary to denary.
 - Subtracting place values for denary to binary.
- ✓ Describe what is meant by the term 'most significant bit'.